

EE1060: Differential Equations and Transform Techniques

Practice Problem Set - 06

Practice problems in topics covered till: Lecture 21

Note : The practice problem sets are designed for your personal review and are not required for submission or evaluation. These problems aim to reinforce your understanding of the material and help you prepare effectively. Some of the practice problems are sourced from the course references, and it is strongly recommended that you consult these references for additional practice problems. Due credit is also given to the TAs of the course, who have contributed to the creation of this problem set.

1. Show that the trigonometric Fourier series of an odd function contains only sine terms.
2. Using the complex form of the Fourier analysis equation, demonstrate that the complex Fourier coefficients of an even function are purely real. **Hint:** An even function can be expressed $g(x) = \frac{1}{2} [g(x) + g(-x)]$.
3. Using the complex form of the Fourier analysis equation, show that the complex Fourier coefficients of an odd function are purely imaginary. **Hint:** Any odd function can be written as $g(x) = \frac{1}{2} [g(x) - g(-x)]$.
4. Compute the Trigonometric Fourier coefficients for the function whose graph is shown in Fig. 1(a).
5. Compute the Trigonometric Fourier coefficients for the function whose graph is shown in Fig. 1(b).
6. Compute the Complex Fourier coefficients for the function whose graph is shown in Fig. 1(c).

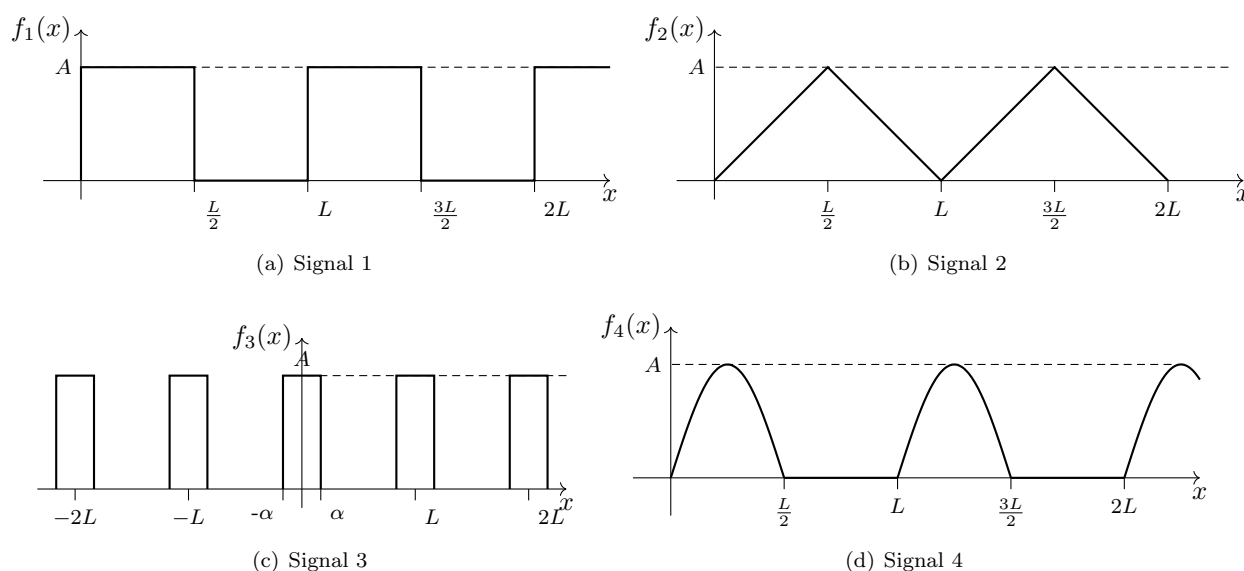


Fig. 1: Waveforms for analysis

7. Compute the Trigonometric Fourier coefficients for the function whose graph is shown in Fig. 1(d). Note that, the function $f_4(x)$ is given by

$$f_4(x) = \begin{cases} A \sin(\omega_0 x) & 0 \leq x \leq \frac{L}{2} \\ 0 & \frac{T}{2} < t < L \end{cases} \quad \text{where } \omega_0 = \frac{2\pi}{L} \quad (1)$$

8. The **Legendre polynomials** $p_n(x)$ are a sequence of orthogonal polynomials that satisfy Legendre's differential equation. The variable n is typically referred to as the degree or order of the polynomial. The first and second degree Legendre polynomials are defined as $p_0(x) = 1$, $p_1(x) = x$. The Legendre polynomials of degree 2 and higher can be derived using the following recurrence relation

$$(n+1)p_{n+1}(x) = (2n+1)xp_n(x) - np_{n-1}(x) \quad (2)$$

- (a) Show that the first three Legendre polynomials $p_0(x)$, $p_1(x)$ and $p_2(x)$ are orthogonal on the interval $[-1, 1]$.
 - (b) Let $f(x)$ denote the projection of $g(x) = x^3$ on the orthogonal set $\{p_0(x), p_1(x), p_2(x)\}$. Compute $f(x)$.
 - (c) Let $h(x)$ denote the projection of $g(x) = x^3$ on the orthogonal set $\{p_0(x), p_1(x), p_2(x), p_3(x)\}$. Compute $h(x)$.
9. Determine if the set $\{1, x, x^3, x^5\}$ is orthogonal on the interval $[-1, 1]$. If so, compute the projection of the function $g(x) = x^4$ onto this set in the same interval.

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